ChatGPT

Functional Requirements

Multi-Role Access & Portals

- **Role-based portals:** Support separate interfaces for landlords (owners), property managers, and tenants. For example, Buildium provides an **Owner Portal** for property owners to view financial reports and a **Resident Center** for tenants to pay rent and submit requests [1]. AppFolio similarly offers *Owner*, *Vendor*, and *Resident* portals and communication templates (email/SMS) to streamline interactions [2]. Administrative roles can manage accounts, user permissions and global settings.

Property Listing & Inventory Management

- **Listing creation and editing:** Allow creation and editing of rental listings with rich content (descriptions, amenities, high-resolution photos and virtual tours). Platforms support multiple media uploads and SEO-friendly fields. Zillow's Rental Network feed, for instance, accepts multiple photos, virtual tours and detailed pricing/availability data [3].
- **Syndication & syndication management:** One-click publishing of listings to partner networks and ILS sites. For example, Realtor.com's Avail platform lets landlords "publish your listing to 20 of the most popular listing sites with a single click" [4]. Buildium also supports syndication to major sites via its **Rental Listings** dashboard [5].
- **Availability management:** Calendar-based inventory tracking. Managers mark unit status (available, leased, pending) and block dates for maintenance. Borrowing from Airbnb's model, hosts can *"open more dates for guests to book and block nights you can't host"* directly on a shared calendar [6].
- **Property metadata:** Store unit details (bed/bath count, square footage, rent amount, pet policies, amenities, location coordinates) and support map API integration (e.g. Google/Bing Maps) to show property location.

Rental Application & Screening

- **Online applications:** Tenants can submit rental applications and pay application fees directly through the platform. For example, Realtor.com/Avail enables renters to *apply directly from your listing on partner sites* to accelerate the lead-to-application process [7]. Buildium lets managers accept applications via the marketing website and process fees online [8].
- **Applicant tracking:** Track each applicant's status (submitted, reviewed, approved/denied) and attach documents. Systems can automate questionnaire forms.
- **Tenant screening:** Integration with background-check services for credit, criminal, eviction history. Buildium's **Tenant Screening** feature runs comprehensive reports on applicants [9]. RentCafe websites similarly allow prospects to complete required screenings during online application, with results syncing to management software [10].
- **Pre-qualification workflows:** Tools to automatically disqualify or flag applicants based on credit scores or income ratios. Some platforms (e.g. integrations with services like Tenant Turner) can pre-screen leads before scheduling showings [11].

Lease Agreements & Renewals

- **Lease document generation:** Provide templated, customizable lease agreements. Buildium and Avail include lawyer-reviewed lease templates. For example, Avail advertises **"Lawyer-Reviewed Leases"** and lets users prepare leases online [12].
- **E-signature and electronic storage:** Full digital signing of leases and renewals. Buildium's **eSignature for Leasing** lets managers send leases online and collect tenant signatures on any device [13]. Signed leases are stored and linked to tenant accounts.
- **Renewals management:** Automated lease renewal reminders and self-renewal options. Systems can notify managers/landlords of upcoming lease expirations and facilitate lease extensions or rent adjustments.
- **Document management:** Central storage of all lease documents and tenant files (IDs, proof of income, correspondence) with version control.

Payments & Accounting

- **Online rent collection:** Enable tenants to pay rent electronically (credit card, ACH/eCheck, digital wallets). Buildium allows managers to *"make and accept payments online for rent, association fees, and vendors"* [14]. AppFolio similarly supports online payments tied to each tenant ledger [15]. Recurring payments and autopay help automate monthly rent collection.
- **Ledgers and bookkeeping:** Automatic posting of all transactions (rent, fees, expenses) to property and company ledgers. Support multi-property consolidation. Generate income statements, balance sheets, and rent rolls.
- **Expense management:** Track maintenance and operating expenses; allocate costs per property. AppFolio and Buildium integrate property accounting (bank reconciliations, GL) into the system. For example, Buildium's **Property Accounting** module secures bank accounts and creates reports [16].
- **Tax compliance:** Automated 1099 generation for vendors. Buildium's **1099 eFiling** feature enables filing 1099s online during tax season [17].
- **Payment portals for owners:** Owner portal dashboards show income/expense reports and distributions.

Maintenance & Service Requests

- **Resident maintenance portal:** Tenants can submit work orders via an online portal or mobile app [18]. They describe issues, attach photos, and track status. Buildium's Resident Center lets tenants **"wrangle maintenance requests"** online [18]. AppFolio similarly includes *"Online Maintenance Requests"* under its maintenance tools [19].
- **Work order management:** Property managers can create, assign, and monitor work orders. The system can assign vendors/contractors, set priorities and due dates. Buildium enables tracking of work orders and paying vendor bills through the platform [20]. AppFolio includes a *"Work Order Management"* module for streamlined scheduling of repairs [21].
- **Unit turnover & inspections:** Scheduling unit turns (vacancy prep) and inspections. AppFolio provides a *"Unit Turn Board"* and mobile inspection tools [21]. Buildium offers built-in inspections (via HappyCo partnership) from mobile devices [22].
- **Preventive maintenance:** Recurring maintenance scheduling (e.g., HVAC service reminders). Noted on master calendar or automated tasks.

- **Maintenance history:** Log all past requests/repairs per unit to inform future scheduling and budgeting.

Communication & Messaging

- **In-app messaging:** Built-in email/SMS templates and messaging between landlords/managers and tenants or vendors. AppFolio highlights “Email and Text Templates” for notifications and enables direct chat via owner/resident portals [2] . Systems often include bulk email or SMS blasts for announcements.
- **Notifications & alerts:** Automated reminders (rent due, lease expiration, maintenance updates). Quick-reply automation (like Airbnb’s templates) can answer common questions or send scheduled messages (Airbnb’s host app lets hosts *“schedule messages to be sent at key moments”* [23]).
- **Lead communication:** Platforms can route rental inquiries as leads; Zillow and Realtor APIs help push leads to CRMs. For example, Zillow’s Rental Network offers a **Lead API** to deliver leads to management systems [24] . Avail allows applicants to apply on partner sites, funneling leads directly to the landlord.

Analytics & Reporting

- **Occupancy and financial dashboards:** Real-time overview of portfolio health (occupancy rate, delinquency rate, rent roll, profit/loss). Buildium offers an **Analytics & Insights** dashboard for data-driven decisions [25] .
- **Custom reports:** Generate reports on income/expenses by property, owner statements, vacancy statistics, late payments, etc. AppFolio provides flexible reporting tools (e.g., leasing metrics dashboard, performance insights) [26] .
- **Business intelligence:** Advanced metrics like NOI, cash flow projections, unit economics. Ability to export data or connect to BI tools via API.
- **Audit trails:** Keep logs of all user actions for security and compliance. AppFolio’s enterprise plans even list “Audit Logs” for tracking system changes [26] .
- **Owner/Investor reporting:** Detailed statements for property owners, possibly with mobile access.

Integrations & APIs

- **Listing networks:** Data feed APIs to syndicate listings. For instance, property management systems often integrate with Zillow’s Rental Network via a standardized feed [3] . This automated feed can submit listing data (photos, pricing, availability) to Zillow at no cost.
- **Payment processors:** Integrations with gateways (e.g. Stripe, Authorize.Net) and ACH networks for secure payment processing. The system should be PCI-compliant for credit card storage.
- **Screening services:** Connectivity to credit bureaus or background-check providers. Buildium partners with screening services (e.g. TransUnion) to automate tenant vetting.
- **Mapping and GIS:** Embed map APIs (Google Maps, Bing Maps) for property location and neighborhood data. This supports features like proximity search and map-based browsing of listings.
- **Accounting software:** Export or sync data to QuickBooks, Xero, etc., if needed for corporate accounting.
- **Third-party services:** Calendar sync (e.g. Google Calendar), email/SMS providers (Twilio, SendGrid), helpdesk systems.

- **Open APIs:** Provide public REST APIs or webhooks so customers can build custom integrations. For example, Buildium's **Open API** exposes property, resident, and accounting data for integration [27]. AppFolio offers APIs in higher-tier plans (read-only in Plus, read/write in Max) [28] [29].
- **Channel management:** For owners who also list short-term, integrate with Airbnb/VRBO to pull availability and bookings if applicable.
- **Smart home / IoT:** Optionally integrate smart locks, thermostats or sensors (e.g. for vacation rentals or upscale properties).

Long-Term Booking & Review Features (BNB-Inspired)

- **Calendar-based availability:** Borrowing from vacation rental systems, each unit has an interactive calendar. Hosts can "block" dates or open availability, set minimum/maximum lease terms, and manage special periods (e.g. blackout for maintenance) via a calendar view [6]. Dynamic pricing suggestions (e.g. discounts for longer stays) can be offered similarly to nightly price tips on Airbnb.
- **Booking workflow:** Even for monthly leases, support an inquiry/booking workflow. Prospective tenants can request a booking or tour and have it automatically logged in the system. As on Airbnb, incoming booking requests (with dates and guest count) are prominently displayed in the host's dashboard [23]. Managers can set "Instant-Book" rules (auto-approve under criteria) or manually vet requests.
- **Reviews/ratings:** Implement a two-sided review system. After move-in or checkout, tenants can rate landlords/properties and vice versa. This builds trust and accountability. Airbnb's host tools include an *"Insights"* section where hosts review guest ratings; similarly, a rental platform can expose tenant ratings and property feedback to future renters [30].
- **Flexible leases:** Support mid-lease edits (e.g. early termination, subletting permissions) and addenda. Unlike fixed-term BNB stays, long-term systems must handle things like lease extensions and leasebreak notices in workflow.
- **Amenities/House rules display:** Show policies (pet rules, smoking) and services (parking, laundry) clearly on each listing, as vacation sites do.

Non-Functional Requirements

- **Scalability:** The system must support hundreds of thousands of listings and concurrent users. Architecture should allow horizontal scaling (e.g. via cloud-based microservices) to handle peak loads (e.g. month-start rent payments, listing launches).
- **Performance:** Fast response times for search queries and UI actions. Listings and photos should load quickly (page load < *2s* for 90% of users). Use efficient databases (e.g. indexed searches, caching) for property and analytics data.
- **Availability & Reliability:** High uptime (≥99.9%). Design for fault tolerance and automated failover. Critical services (payments, communication) should have redundancy. Disaster recovery plans and regular backups are required.
- **Security & Compliance:** Enforce strong authentication (multi-factor admin login) and fine-grained RBAC so users see only their data. Encrypt sensitive data at rest (tenant PII, payment info) and in transit (TLS). Comply with privacy regulations (GDPR/CCPA) for personal data. For payments, meet PCI-DSS standards. Follow fair housing and landlord-tenant law compliance (data retention, disclosures). Implement audit logging of user actions.
- **Usability & Accessibility:** Intuitive, responsive UI for web and mobile. Ensure mobile-friendly design (many tenants use smartphones). Provide accessibility (WCAG) features. Simple workflows (e.g. one-

click listing syndication) improve adoption. Provide multi-language/currency support if multi-national.

- **Maintainability:** Modular, well-documented codebase. Use CI/CD pipelines and automated testing for reliable updates. Design APIs and data models to be extensible (e.g. adding new property types or fees).
- **Data Integrity & Privacy:** Regular data backups and integrity checks. Tenants' and owners' personal information must be protected and segregated.
- **Monitoring & Analytics:** Built-in logging, alerts, and health dashboards for system metrics. Track error rates, API performance, and usage patterns to scale capacity as needed.

Sources: Feature lists and developer guides from production platforms. For example, Zillow's developer portal documents listing feed capabilities [3]; Realtor.com/Avail marketing highlights listing syndication, tour scheduling, applications, and lease generation [4] [7] [12]; Buildium and AppFolio official sites enumerate leasing, payments, maintenance, portals, analytics and APIs [1] [2] [25] [28]. RentCafe/Yardi materials describe online applications and site integration [31], and Airbnb's own documentation illustrates calendar management and review features [6] [32]. These collectively inform the above requirements.

---

[1] [5] [8] [9] [11] [13] [14] [16] [17] [18] [20] [22] [25] [27] Property Management Software Features | Buildium
https://www.buildium.com/features/

[2] [15] [19] [21] [26] [28] [29] Property Management Software Features | Book a Free Demo
https://www.appfolio.com/property-manager

[3] Rentals Feed Integrations - Data & APIs
https://www.zillowgroup.com/developers/api/rentals/rentals-feed-integrations/

[4] [7] [12] Rental Listings: Get More From Your Online Listings | Avail | realtor.com
https://www.realtor.com/landlords/list-rental-property

[6] [23] [30] [32] New hosting tools to help grow your business - Resource Center - Airbnb
https://www.airbnb.com/resources/hosting-homes/a/new-hosting-tools-to-help-grow-your-business-738

[10] [31] RentCafe websites are a property manager's best friend
https://www.yardibreeze.com/blog/2025/04/rentcafe-websites/

[24] Lead API - Data & APIs
https://www.zillowgroup.com/developers/api/rentals/lead-api/